Discrete logarithm

Module I

Logarithm

Can you Compute

$2^5 = 32$

Can you compute $3^? = 129140163$

Answer – 17

We can also write

$5 = \log_2(32)$ $17 = \log_3(129140163)$

In general

$p^i = q$

$i = \log_p(q)$

Modular Arithmetic

Can you compute this

$\mathbf{9} = 3^4 \bmod 12$

$26067 = 14^i \pmod{29867}$

$26067 = 14^{12} \pmod{29867}$

$4 = \mathrm{dlog}_{3,12}(9)$

$12 = \mathrm{dlog}_{14,29867}(26067)$

In General

$a = b^i \pmod{m}$

$i = \mathrm{dlog}_{b,m}(a)$

More Examples

- Compute

$$2^5 \bmod 3 = 2$$

$$4^4 \bmod 11 = 3$$

Can you compute this ?

$8 = 5^i \bmod 13$ - determine i

or $i = dlog_{5,13}(8)$

Ans – $3 = dlog_{5,13}(8)$ or

$$8 = 5^3 \bmod 13$$

Check this one

$? = 5^7 \bmod 13$ and $? = 5^{11} \bmod 13$

- **We should consider cases only when dlog gives us a unique answer**
- **Only then we can use dlog in our algorithms**

Primitive Roots

unique

$a = b^i \pmod m$

b should be primitive root of prime m

$i = \mathrm{dlog}_{b,m}(a)$

We can get an unique value of i if b is a **primitive root** of **prime modulo** m

What is a primitive root ?

Primitive Root

$b^i \bmod m$

1 mod 13 $\quad a = b^i \pmod{m}$ (consider $m=13$) $\quad b^{m-1}$

b	b^1	b^2	b^3	b^4	b^5	b^6	b^7	b^8	b^9	b^{10}	b^{11}	b^{12} mod m
1	1	1	1	1	1	1	1	1	1	1	1	1
2	2	4	8	3	6	12	11	9	5	10	7	1
3	3	9	1	3	9	1	3	9	1	3	9	1
4	4	3	12	9	10	1	4	3	12	9	10	1
5	5	12	8	1	5	12	8	1	5	12	8	1
6	6	10	8	9	2	12	7	3	5	4	11	1
7	7	10	5	9	11	12	6	3	8	4	2	1
8	8	12	5	1	8	12	5	1	8	12	5	1
9	9	3	1	9	3	1	9	3	1	9	3	1
10	10	9	12	3	4	1	10	9	12	3	4	1
11	11	4	5	3	7	12	2	9	8	10	6	1
12 (m-1)	12	1	12	1	12	1	12	1	12	1	12	1

Definition

Primitive Root

- If b is a primitive root of mod m where m is a prime number then b^1 mod m ,b^2 mod m ,b^3 mod m ..b^{m-1} mod m are distinct values

Discrete Logarithm –

If a is an arbitrary integer relatively prime to m and b and is a primitive root of m , then there exists exactly one number i such that

$$a = b^i \pmod{m}$$

The number i is then called the discrete logarithm of a with respect to the base b modulo m and is denoted

$$i = \mathrm{dlog}_{b,m}(a)$$

Check if 5 is primitive root of mod 7?

Exercise

- Check if 5 is a primitive root of mod 7 ?

$i = \text{dlog}_{b,m}(a)$ or $a = b^i \pmod{m}$ tough

Determine i when a b and m are given
Finding Discrete Logarithm for Large Numbers will be difficult

Exercise

- $? = \text{dlog}_{17,2111}(1922)$
- $1922 = 17^i \pmod{2111}$? 2048 bits

Ans = 12